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Original Article

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A longitudinal assessment of sleep architecture in children and adolescents with craniopharyngioma

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Abstract

Study Objectives: Craniopharyngioma is a suprasellar brain tumor often associated with hypersomnia, which may be potentially due to alterations in sleep architecture. We performed cross-sectional and longitudinal assessments of sleep architecture in pediatric patients with craniopharyngioma.
Methods: We evaluated 94 patients (median age 9) enrolled in a clinical trial that included limited surgery with proton radiotherapy or radical surgery with observation. Nocturnal polysomnography and multiple sleep latency testing were performed at baseline and at 36 months after enrollment. Sleep architecture was compared to that of healthy children reported by Zhu et al., as well as longitudinally and across hypersomnia diagnostic groups using standard median comparison tests. Chi-squared tests assessed longitudinal changes in prevalence of hypersomnia disorders.
Results: Pediatric patients with craniopharyngioma exhibited 1.0% and 1.5% lower N1 and 9.21% and 9.59% lower N2 sleep ($p < .001$) at baseline and 36 months than healthy counterparts. N3 sleep was elevated by 11.84% and 13.47% respectively ($p < .001$). At 36 months, 2.63% less time was spent in REM ($p = .005$). Analysis of the entire cohort revealed no statistically significant longitudinal changes in sleep architecture ($p > .05$). However, in a subset of patients with advancing Tanner stage, N1 sleep decreased and REM sleep increased over 36 months. There were no longitudinal changes in prevalence of hypersomnia disorders ($p > .05$), nor differences in sleep architecture between patients with and without hypersomnia.
Conclusions: Children and adolescents with craniopharyngioma exhibit atypical sleep architecture following surgery, that persists with long-term follow-up. Further research is needed to determine if sleep architecture mediates hypersomnia.

Key words: craniopharyngioma; pediatric sleep architecture; hypersomnia; narcolepsy type 2

Statement of Significance

Hypersomnia disorders are associated with altered sleep architecture in individuals without craniopharyngioma. Disruptions in sleep architecture can contribute to memory impairment, a long-term morbidity following treatment for craniopharyngioma. This study is the first to document atypical sleep architecture in pediatric patients with craniopharyngioma.

Introduction

Craniopharyngiomas are non-malignant, locally invasive tumors arising in the suprasellar region [1]. Both rare and highly treatable, they comprise only 5%–10% of pediatric brain tumors, with an incidence of 0.19 per 100,000 persons in both adults and children and 10-year overall survival rates of 86%–99% [2–4]. Tumor growth can be aggressive, often resulting in damage to optic pathways, the hypothalamus, and other critical structures [1, 5]. Furthermore, common treatment approaches, which include surgery and radiotherapy, can lead to neurologic, endocrine, and cognitive deficits [5–7].

Despite high survival rates, children with craniopharyngioma are at elevated risk for significant long-term morbidities related to both tumor and treatment [7, 8]. Such morbidities include

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cognitive impairment, pituitary deficiencies, hypothalamic dysfunction, and fatigue [7–9]. Deficits in working and episodic memory are of particular concern; their cause, though associated with hypothalamic involvement (HI), remains undefined [7, 10, 11]. HI is associated with sleep–wake dysregulation and the development of hypersomnia, including narcolepsy due to medical condition without cataplexy and hypersomnia due to medical condition [9, 12–14]. Patients without craniopharyngioma but with narcolepsy have lower health-related quality of life and more challenges with school functioning than heathy counterparts do [15, 16]. Further research is needed to fully establish the impact of hypersomnia disorders on daytime functioning in craniopharyngioma survivors [9, 17].

Although studies associate HI, hypersomnia disorders and memory deficits, further research is needed to determine the underlying mechanisms [7]. Sleep architecture, the structure of sleep throughout the major sleep period, may be a promising path of exploration [18]. Sleep physiology is characterized by four distinct stages based on oscillatory patterns in electroencephalography, eye movements measured with electro-oculogram, and muscle tone measured with submental muscle tone [19]. N1 sleep or light sleep involves primarily theta frequency oscillations, N2 sleep involves deeper sleep and is characterized by the emergence of K-complexes and sleep spindles, N3 or slow wave sleep is characterized by high amplitude slow wave oscillations, and REM (rapid eye movement sleep) is characterized by mixed theta and slow alpha oscillations and "sawtooth waves" on EEG, muscle atonia and phasic rapid eye movements [20]. Disruptions in sleep architecture, often accompanied by sleep fragmentation, are well established as a cause of or contributor to memory deficits [18]. Emerging research suggests that EEG oscillation characteristics of both REM and non-REM sleep, which underlie the expected changes between sleep stages and contribute to normal sleep architecture, are critical for memory consolidation [21, 22].

Altered sleep architecture may be associated with hypersomnia disorders. Previous studies indicate that patients with narcolepsy type one (NT1), a disorder due to the loss or dysfunction of neurons in the hypothalamus, display altered sleep architecture with either an increase in N1 sleep or an increase in both N3 and REM [23–25]. Three case studies have evaluated sleep architecture in patients with craniopharyngioma, who are at risk for hypersomnia disorders: two of three subjects met criteria for narcolepsy (one with narcolepsy due to medical condition with cataplexy, one with narcolepsy due to medical condition without cataplexy) [26, 27]. The latter patient displayed an abnormally elevated percentage of REM sleep; the former patient had no abnormalities in sleep architecture [26, 27]. The third subject experienced reduced N2 and N3 sleep and subjective hypersomnia but did not meet narcolepsy diagnostic criteria [26]. These limited findings suggest that both the tumor and treatment may disrupt sleep architecture and that these alterations may differ between hypersomnia diagnostic groups.

Despite significantly elevated risk for hypersomnia disorders and cognitive deficits in patients with craniopharyngioma, no research—aside from the previous case studies—has assessed sleep architecture in this cohort and whether it differs from sleep architecture in healthy children and adolescents. Therefore, we aimed to longitudinally assess sleep architecture in patients with craniopharyngioma in comparison to that of healthy children and adolescents. Furthermore, we sought to identify differences in sleep architecture according to the presence or absence of a hypersomnia disorder. We hypothesized that patients with craniopharyngioma would exhibit greater N1 and REM sleep than their healthy counterparts and that patients with hypersomnia would display greater N1 sleep than would those without it.

Materials and Methods

Participants

One hundred ten patients with craniopharyngioma were enrolled in a Phase II trial of proton therapy (ClinicalTrials.gov Identifier NCT01419067) following limited surgery when indicated. Informed consent and assent were obtained for all protocol-related procedures. Patients aged 0–21 years with a diagnosis of craniopharyngioma confirmed by intraoperative findings or neuroimaging were eligible for participation. The study was approved by the Institutional Review Board of St. Jude Children's Research Hospital.

Study design

Participants ages 2–20 years of age with residual tumor after surgery were treated with proton therapy and observed for 60 months. Patients without residual tumor tissue were observed for 60 months. Participants completed clinical sleep assessments at baseline (before the initiation of any indicated proton therapy) and at 36 months (±90 days) after commencement of therapy or observation. Clinical sleep assessments consisted of an in-person evaluation by a board-certified pediatric sleep medicine physician, nocturnal polysomnography (PSG), and a multiple sleep latency test (MSLT) the next day. All sleep assessments were completed at an American Academy of Sleep Medicine (AASM)-accredited sleep center using AASM recommended protocols, including EEG derivations from frontal, central, and occipital scalp regions in each participant [19]. Patients younger than 6 years were excluded from the MSLT. Other clinical variables (BMI z-score, Tanner stage) were collected at same-visit endocrine appointments. Preoperative HI, a measure of damage to the hypothalamus, was also assessed via neuroimaging and scored as no HI (grade 0), anterior HI (grade 1), or anterior and posterior HI with mammillary bodies included (grade 2) [28].

Measures

Polysomnography. An attended nocturnal PSG commenced within 1 h of the participant's habitual bedtime. During PSG, two-channel electro-oculogram, one-channel submental electromyogram, and electroencephalogram (incorporating bilateral frontal, central and occipital derivations) were used to characterize sleep–wake state and sleep stages. Respiratory effort at chest and abdomen, oronasal thermistor and nasal pressure transducer–measured airflow, continuous oximetry, electrocardiogram and audio recordings detecting snoring and other sounds of obstruction were used to assess sleep-related breathing. Finally, positional sensors, electromyogram or anterior tibialis accelerometry, and videotape documented limb movements and other sleep-related behaviors. Sleep stages were identified and sleep–wake scoring was performed using standard AASM scoring criteria and techniques including pediatric scoring guidelines unless the participant was 18 years of age or greater, in which case adult scoring criteria were used [19]. REM latency was not assessed.

Multiple sleep latency test. Patients aged 6 years or older completed the MSLT, an in-laboratory, standardized measure of tendency for sleepiness. MSLTs were performed according to AASM guidelines. The MSLT consists of four or five 20-min nap opportunities in a darkened room at 2-h intervals. If patients entered

sleep during a nap opportunity, they were allowed 15 min of sleep time before being awakened by staff. If no sleep occurred within 20 min, the nap opportunity ended. Mean sleep latency (MSL) and sleep stages were recorded for each nap opportunity. Criteria for hypersomnia based on MSLT findings is classified according to pubertal status. For pre-pubescent, Tanner stage 1 patients, hypersomnia was defined as a MSL value of ≤15 min; for pubescent (Tanner stage 2 or greater), MSL ≤ 10 min fulfilled MSLT hypersomnia requirements [29]. More specifically, narcolepsy due to a medical condition was defined as meeting the MSLT criteria for hypersomnia for pubertal stage and the presence of sleep-onset REM entries during two or more nap opportunities [14]. Hypersomnia due to a medical disorder was defined as meeting the MSLT criteria for hypersomnia for pubertal stage and the presence of fewer than two naps with sleep-onset REM episodes [14].

Statistical analysis

Descriptive statistics were performed for all demographic, clinical, and sleep variables at the baseline and 36-month time points. Sleep variables at each time point were compared to those of healthy children and adolescents from Zhu et al. [30]. Sleep variable data at each time point were first assessed for normality using Q-Q-plots and Shapiro–Wilk tests. Sleep variables determined to be approximately normal were then compared to healthy children using one-sample *t*-tests, and non-normal variables were tested using Wilcoxon signed-rank tests. Multiplicity across tests was controlled using the Benjamini–Hochberg false discovery rate (FDR), and both raw *p*-values and FDR-adjusted *q*-values are reported.

Zhu et al. assessed 118 healthy children and adolescents ages 5-18 (Mean age = 13.1 ± 3.3; BMI z = 0.85 ± 1.00; 44.9% male; 74.6% pubertal) who underwent one night PSG. PSG was scored according to AASM criteria by a sleep technologist [30].

Comparisons between time points used Wilcoxon rank-sum tests or independent sample t-tests for unpaired data and Wilcoxon signed-rank tests or paired *t*-tests for paired data. Differences between sleep architecture variables according to sleep diagnosis were assessed via Kruskal-Wallis tests and a one-way ANOVA according to data normality. A two-sided *p*-value <0.05 was used to determine statistical significance for all tests. All analyses were performed by using R (Version 4.40).

Results

Demographic and clinical characteristics

Of 110 participants, 94 completed a baseline PSG, and 80 completed a 36-month follow-up PSG. The participants were predominantly female (54% (Baseline), 56% (36 Months)), white (74% (Baseline), 75% (36 Months)), and overweight or obese (mean BMI-Z of participants <21 = 1.35 ± 1.02 (Baseline), 1.51 ± 0.94 (36 Months)). 13 participants had grade 0 HI, 25 participants had grade 1 HI, and 56 participants displayed grade 2 HI—the most severe level of damage to the hypothalamus. Of the 94 patients who completed baseline assessments, 80 also completed next-day MSLT (mean age, = 10.63 ± 3.89 years). Similarly, at the 36-month follow-up, 65 of 80 participants completed next-day MSLT (mean age = 12.85 ± 4.54 years) (see Table 1).

Sleep architecture comparisons between patients with craniopharyngioma and healthy children and adolescents

Baseline. Children and adolescents with craniopharyngioma spent 1.0% less of their total sleep time in N1 than healthy children and adolescents ($p < .001$) (see Table 2) [30]. Furthermore, the percentage of time spent in N2 by patients with craniopharyngioma was significantly less than their healthy counterparts, with a difference in means of 9.21% ($p < .001$). In contrast, the percentage of N3 sleep was 11.84 percentage points greater in children and adolescents with craniopharyngioma than those without ($p < .001$). There was also a marginally significant difference in percentage of total sleep time spent in REM: children and adolescents with craniopharyngioma spent 1.7% less sleep time in REM ($p = .056$) than healthy counterparts. The sleep efficiency (SE) of children and adolescents with craniopharyngioma exceeded the SE of healthy controls by 5.01% ($p < .001$). Finally, children with craniopharyngioma, on average, slept 36.43 more minutes than the healthy comparators ($p < .001$) (see Table 2).

36-month follow-up. At the 36-month follow-up sleep study, the percentage of N1 sleep in both children and adolescents with craniopharyngioma was lower than that in their healthy counterparts by 1.55 percentage points ($p < .001$). Similarly, the percentage of time spent in N2 was significantly lower in children and adolescents with craniopharyngioma by 9.59 percentage points ($p < .001$). Patients with craniopharyngioma also had a significantly greater percentage of N3 sleep, with median differences of 13.47% ($p < .001$). Furthermore, children and adolescents with craniopharyngioma spent 2.63% less of their total sleep time in REM than healthy children and adolescents at the 36-month follow-up ($p = .005$). SE was also significantly greater in children and adolescents with craniopharyngioma, with 5.78% more time in bed spent in sleep ($p < .001$). Lastly, children and adolescents with craniopharyngioma spent 28.34 more minutes asleep, on average, than their healthy counterparts ($p < .001$) (see Table 2).

Longitudinal analyses

Longitudinal changes in sleep architecture. Following analysis of both patients with data at both timepoints and analysis of all patients, there were no statistically significant changes in sleep architecture variables from the baseline to the 36-month follow-up point ($p > .05$) (Supplementary Table 1). As expected due to pubertal delays associated with craniopharyngioma [7], only 31 participants displayed sexual maturation with changes in Tanner stage from the baseline to the 36-month time point. In a separate analysis of these participants with sexual maturation, N1 sleep decreased by 1 percentage point ($p = .02$), and REM sleep increased by 4.7 percentage points ($p = .012$) (Figure 1). N2, N3, WASO, or SE were statistically similar at both time points for this subgroup ($p > .05$) (Figure 1 and 2).

Longitudinal changes in sleep diagnosis prevalence. Participants without MSLT data were excluded from this analysis. For the combined sample of children and adolescents who had completed MSLTs (Baseline, $n = 80$; 36 months, $n = 65$), there were no significant differences in the prevalence of a diagnosis of hypersomnia due to a medical condition or narcolepsy due to a medical condition from baseline to 36-month follow-up sleep study ($p > .05$) or in the prevalence of the combined group (those with either type of hypersomnolence) from baseline to 36-month follow-up ($p > .05$) (see Figure 3).

Differences in sleep architecture by sleep diagnosis

There were no significant differences ($p > .05$) in any sleep architecture variable, (N1, N2, N3, REM) nor in SE or WASO between

Table 1. Participant demographics

Characteristic	Baseline (*n* = 94)	36 months (*n* = 80)
Age (Q1, Q3)	9 (7, 13)	12 (9, 16) years
Sex, *n* (%)	51 (54%)	45 (56%)
Race, *n* (%)	70 (74%)	60 (75%)
BMI-Z	1.51 (0.805, 2.15)	1.68 (1.04, 2.30)
2-10 years old, *n*	57	34
11+ years old, *n*	37	46
Completed both time points, *n*	74	74
Completed MSLT, *n*	80	65
Hypersomnia, *n*	40	36
Narcolepsy, *n*	27	20
No Diagnosis, *n*	13	9
MSLT average sleep onset latency in minutes (Q1, Q3)		
Hypersomnia	7.5 (4.8, 9.8)	4.8 (3.1, 7.3)
Narcolepsy	2.4 (2.0, 5.0)	3.1 (1.5, 6.0)
No diagnosis	18.5 (15.6, 19.9)	15.6 (11.0, 17.9)
MSLT sleep onset REM periods n (Q1, Q3)		
Hypersomnia	0 (0, 1)	0 (0, 0)
Narcolepsy	2 (2, 3)	2 (2, 3)
No diagnosis	0 (0, 0)	0 (0, 2)
Observation, *n*	5	5
Proton therapy, *n*	89	75
Hypothalamic involvement, *n*		
Grade 0	13	
Grade 1	25	
Grade 2	56	

Demographics of participants at baseline and 36 months.

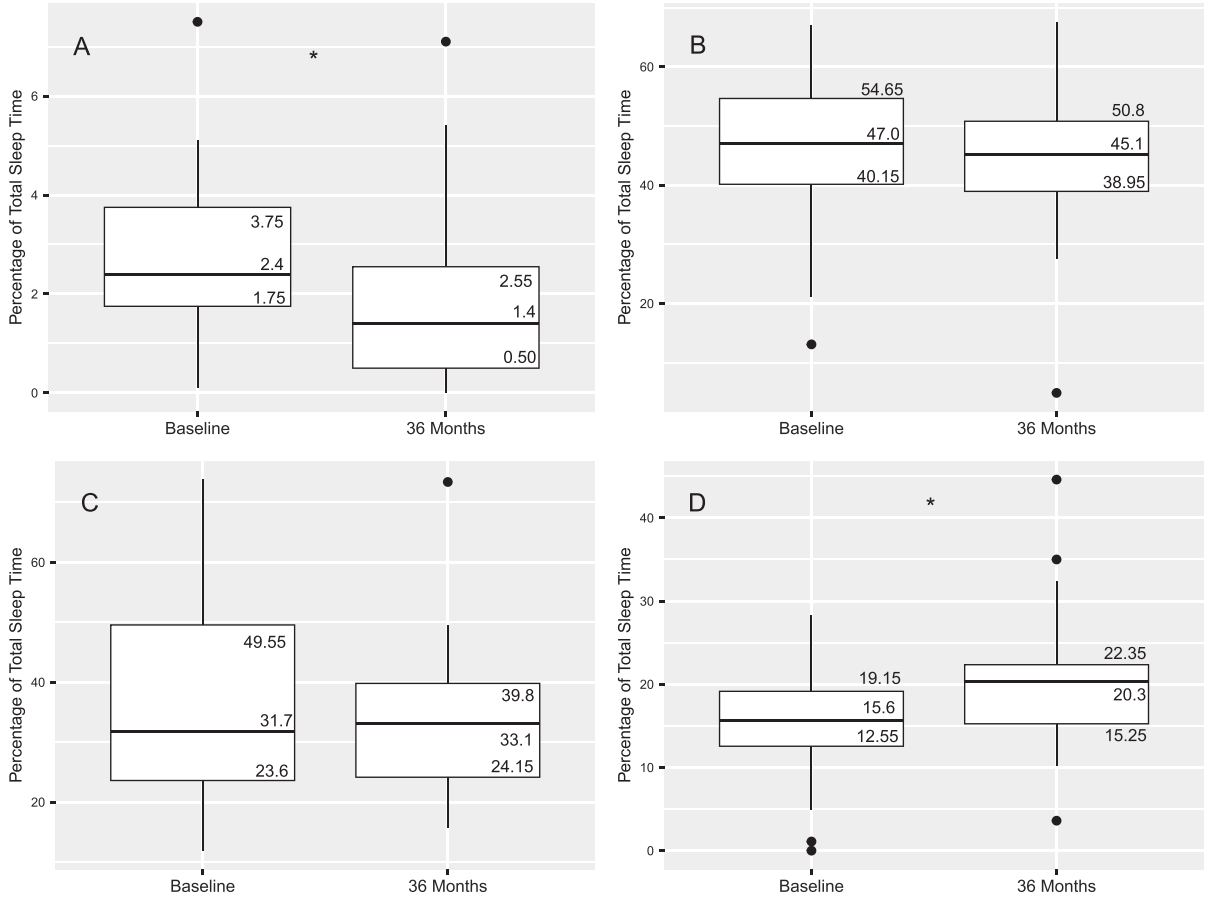

Figure 1. Percentage of total sleep time spent in each sleep phase at both baseline and 36-month follow-up polysomnography in the subset of patients with craniopharyngioma who displayed changes in Tanner stage. (A) Median percentage of N1 sleep. (D) Median percentage of REM sleep. (B, C) median percentages of N2 and N3 sleep. *$p < .05$; baseline vs. 36 months.

Table 2. Sleep architecture in patients ≥5 years with Zhu et al. [30] reference data

Sleep feature	Healthy children and adolescents	Baseline	36-months
N1%	2.9 [1.8, 4.1]	1.9 [0.6, 3.3]†***	1.4 [0.3, 2.9]‡***
N2%	50.9 ± 7.4	41.7 ± 14.0 (38.7–44.7)‡***	41.3 ± 14.7 (38.0–44.6)†***
N3%	25.0 ± 8.4	36.8 ± 16.7 (33.3–40.4)‡***	38.5 ± 17.1 (34.7–42.3)†***
REM%	20.8 ± 5.0	19.1 ± 8.2 (17.4–20.8)‡*	18.2 ± 8.0 (16.4–20.0)†**
SE	85.0 ± 10.4	90.0 ± 6.7 (88.6–91.4)‡***	90.8 ± 7.8 (89.0–92.5)†***
TST	443.0 ± 68.0	479.4 ± 60.6 (466.5–492.4)†***	471.3 ± 64.8 (456.9–485.8)†***

Comparisons between children and adolescents with craniopharyngioma and healthy counterparts. q = FDR-adjusted across all tests. †Observed median [Q1, Q3] for Wilcoxon. ‡Mean ± SD (95% CI) for *t*-tests. *p and $q = .056$. **p and $q = .05$. ***p and $q < .001$.

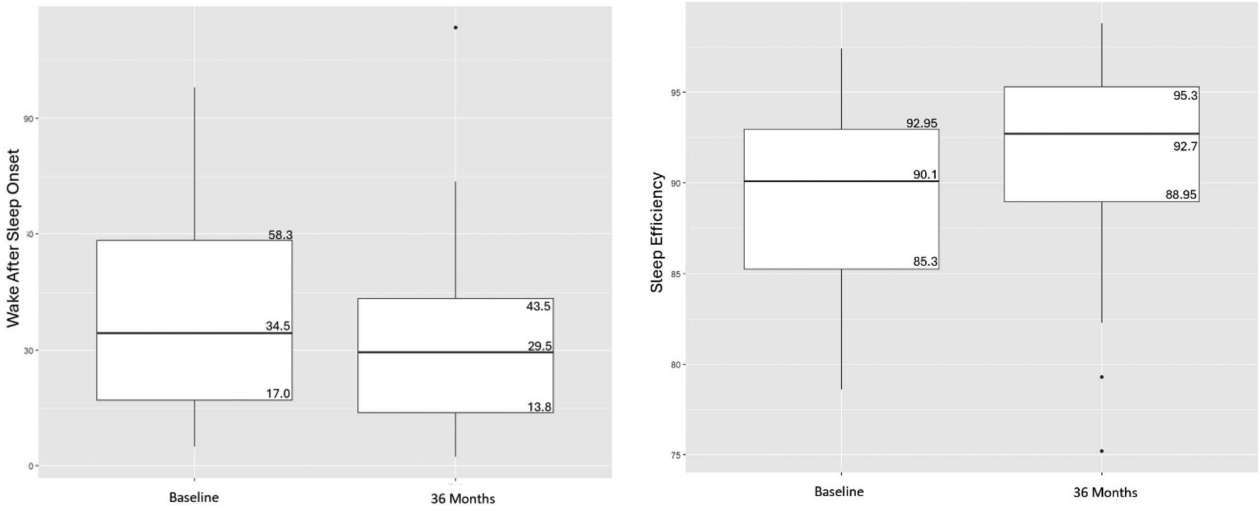

Figure 2. Wake after sleep onset and sleep efficiency in the subset of patients with craniopharyngioma who experienced Tanner stage advancement.

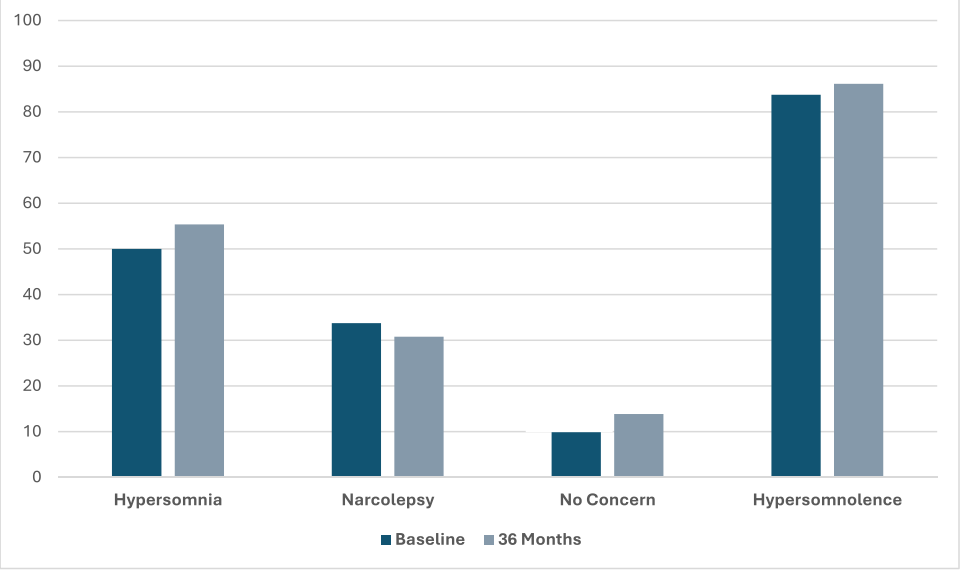

Figure 3. Changes in prevalence of diagnosis from baseline to final time points for the combined age group sample.

patients with hypersomnia due to a medical disorder or narcolepsy due to a medical disorder or those with either form of hypersomnia diagnosis at baseline or 36-month follow up (Supplementary Table 3). Moreover, when patients were collapsed into a single group (referred to in Supplementary Table 4 as "Hypersomnolence"), those with and without disorders of hypersomnolence did not significantly differ at either time point for any sleep variables ($p > .05$).

Discussion

Similar to patients in case studies on craniopharyngioma published by Tachibana et al. and Cordani et al. [26, 27], participants in this study displayed atypical sleep architecture. In each study, the distribution of sleep stages was atypical compared with the sleep architecture of healthy adolescents, with our study finding reduced N1 and N2 sleep, with elevated N3 sleep [26, 27]. This assessment also revealed a reduction in REM sleep at the

36-month follow up as well as marginally less REM sleep at baseline. In comparison to previous literature, our study assessed a much larger sample; therefore, the results may be more generalizable. Patients with craniopharyngioma with and without evidence of excessive daytime sleepiness on the MSLT had similar sleep architecture. These findings suggest that narcolepsy and hypersomnia may occur independently of disrupted sleep architecture. This may suggest that sleep architecture differences in children and adolescents with craniopharyngioma are caused not by the disorders of hypersomnolence themselves, but by an independent pathophysiological event. Damage to the hypothalamus is a plausible explanation for either the hypersomnia, the altered sleep architecture or possibly both. It has been well documented that hypothalamic damage, particularly damage to orexin synthesizing neurons in the lateral hypothalamus, can disrupt sleep wake signaling and REM sleep, leading to narcolepsy and hypersomnia [31, 32]. Over half the patients in this study had grade 2 HI, which emphasizes the importance of further research assessing HI and sleep architecture together. Future research might investigate if changes in sleep architecture and the development of narcolepsy and hypersomnia in this patient population are both independent and interacting signs or symptoms stemming from hypothalamic damage. A further pathway for altered sleep architecture may lie in growth hormone deficiency; while N3 sleep is critical to growth hormone release as well as memory consolidation, patients with craniopharyngioma still experience difficulties in both. Some research, however, has shown that children with growth hormone deficiency may experience elevated levels of N3 sleep and/or depressed levels of N2 sleep, and this, therefore, may offer a partial explanation for alterations in sleep architecture [33].

Multiple studies indicate that sleep architecture changes with age and pubertal status [34–36], but no other studies have assessed longitudinal changes in sleep architecture in the craniopharyngioma patient population. From childhood to adolescence, N1 and N2 sleep are expected to increase, while N3 sleep decreases [34, 35]. In longitudinal studies, REM sleep significantly decreases from early childhood to later childhood, and REM increases slightly from late childhood to adolescence [35, 36]. Our participants displayed altered sleep architecture immediately after tumor resection; these patterns remained altered into adolescence.

Given that only a minority of participants' Tanner Stage progressed, pubertal delays may be a mechanism underlying delays in the development of adult sleep architecture in this population. Participants who experienced sexual maturation between the baseline and 36-month time points displayed longitudinal changes in their sleep architecture. Healthy adolescents have been described as spending more time in N1 and N2 sleep and less time in N3 sleep as they transition to adulthood [35, 36]. In this sample of adolescents with craniopharyngioma, N1 sleep decreased without changes in the percentages of N2 or N3 sleep. REM sleep increased, as in previous literature [36]. Because these findings reflect atypical development of sleep architecture even in patients with more typical pubertal development, they highlight the need for further research to identify the relationship between pubertal status and sleep architecture in patients with craniopharyngioma.

Healthy sleep architecture is critical to memory formation and consolidation. Disruption can cause memory deficits, which are highly prevalent in this population and are also linked to disruptions in endocrine function [18, 33]. Therefore, this first large scale identification of atypical sleep architecture in pediatric patients with craniopharyngioma offers important contributions to creating foundational models of sleep, growth hormone deficiency, pubertal development, and cognitive function in children with complex endocrine and neurological disease and is the basis for much research to come.

Importantly, the current study begins to explore a critical, under-researched area in long-term craniopharyngioma morbidities by using a large sample size for a rare disease. Additional strengths include objective measures of sleep architecture and long-term longitudinal analysis.

This study should also be viewed in context with its limitations. Since only one night of PSG was performed at each time point due to sleep data being a secondary outcome in a larger clinical trial, patients may have exhibited the "first night effect" (FNE) or the "reverse first night effect" (RFNE) [37–39]. However, Zhu et al., the comparator study for healthy children and adolescents, also used first night PSGs [30]. Therefore, we can at least conclude that the first night PSG profile of patients with craniopharyngioma differs significantly from first night PSG profiles in healthy children and adolescents, an important clinical observation for the treatment of sleep challenges in this patient population, particularly when patients cannot be scheduled for repeated PSG [30]. Additionally, as one night PSG was used across all time points, any possible FNE and RFNE should not affect longitudinal within group comparisons. Other factors, however, such as patients' anxiety or stress in the sleep laboratory environment, pain, medical illness, medication effects, and in a few participants, prior sedation for imaging procedures may have impacted sleep architecture. Efforts were made to avoid this latter situation whenever possible. Moreover, because of the small number of observation-only patients, no assessments were performed between patients who underwent proton therapy and those on observation only. Secondly, while this study employed frontal electrodes to help score slow wave sleep, Zhu et al., despite using AASM scoring criteria, did not use frontal electrodes. This opens the possibility for changes in SWS (N3) between patients with craniopharyngioma and those without to be due to differences in measurement rather than differences in condition. Studies that use frontal electrodes show 4.9% higher N2 and 2.4% higher N3 than those that do not use front electrodes. The differences in sleep stages observed in this study, 9.21%, and 9.59% for N2 and 11.84% and 13.47% for N3, are much larger, and therefore unlikely to be accounted for by the difference in methodology [40]. A study that used both AASM scoring criteria and frontal electrodes for a robust sample of children and adolescents could not be identified, pointing to a need for more normative studies of sleep architecture in this developmental period.

Additionally, in the comparison study, Zhu et al., PSGs were scored by a sleep technologist without sleep physician "over-reading," as compared with the sleep physician "over-read" of scoring performed by a sleep technologist in this study [30]. Furthermore, only a small subset of patients had no sleep concerns—there may be significant differences in sleep architecture between sleep diagnostic groups that this sample size could not detect. Sleep architecture assessments occurred only after surgical resection, so changes due specifically to surgery were not identifiable. The sample also had limited racial diversity, and gender identity was not noted.

Although patients with craniopharyngioma have been found to fall into distinct subgroups demarcated by sleep onset latency and WASO, this study did not evaluate these subgroups, which represents an additional study limitation [41]. In Witcraft et al., 69% of patients with craniopharyngioma were classified as "Healthy Sleepers" by a latent profile analysis; 21% were

"Night Wakers" with abnormal WASO, and 10% were "Delayed Sleepers" with elevated SOL [41]. An analysis evaluating sleep architecture in these latent classes in this patient population may help elucidate whether data from prior case studies represent clinically meaningful subgroups or outliers. However, no research has assessed whether sleep architecture profiles similar to those in Witcraft et al. emerge in patients with craniopharyngioma [41].

In healthy individuals, typical sleep architecture is thought to support memory consolidation. In patients with craniopharyngioma, who are at elevated risk for both atypical sleep architecture and memory deficits, assessing sleep architecture may help to identify mechanisms underlying memory deficits and may lead to interventions to improve neurocognitive morbidities. Moreover, as patients with craniopharyngioma are likely to display altered sleep architecture in the absence of diagnosed disorders of hypersomnolence, assessment for sleep architecture disruptions should be considered even in patients without daytime or nocturnal sleep concerns. Furthermore, this study did not explore the relationship between HI and disrupted sleep architecture, nor the relationship between altered sleep architecture and cognitive and psychosocial wellbeing in this patient population. Therefore, future research is critical to correlating sleep architecture and cognitive, physiological, and psychosocial wellbeing in patients with craniopharyngioma.

Supplementary material

Supplementary material is available at *SLEEP Advances* online.

Author contributions

Jordan A. Alexandra Davidson (Formal analysis [equal], Visualization [lead], Writing—original draft [lead], Writing—review & editing [equal]), Dora Obodo (Formal analysis [equal], Visualization [supporting], Writing—review & editing [supporting]), Merrill S. Wise (Conceptualization [supporting], Data curation [lead], Methodology [lead], Writing—review & editing [equal]), Thomas Merchant (Conceptualization [lead], Investigation [lead], Methodology [lead], Resources [lead], Supervision [lead], Writing—review & editing [supporting]), Belinda N Mandrell (Conceptualization [supporting], Methodology [lead], Project administration [equal], Resources [equal], Supervision [supporting], Writing—review & editing [supporting]), Valerie McLaughlin Crabtree (Methodology [supporting], Supervision [lead], Writing—original draft [equal], Writing—review & editing [equal])

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Data availability

All data will be made available upon request.

References

1. May JA, Krieger MD, Bowen I, Geffner ME. Craniopharyngioma in Childhood. *Advances in Pediatrics*. 2006;**53**(1):183–209. https://doi.org/10.1016/j.yapd.2006.04.013
2. Ostrom QT, Cioffi G, Waite K, Kruchko C, Barnholtz-Sloan JS. CBTRUS statistical report: primary brain and other central nervous system Tumors diagnosed in the United States in 2014-2018. *Neuro-Oncol*. 2021;**23**(12 Suppl 2):iii1–iii105. https://doi.org/10.1093/neuonc/noab200
3. Edmonston DY, Wu S, Li Y, Khan RB, Boop FA, Merchant TE. Limited surgery and conformal photon radiation therapy for pediatric craniopharyngioma: long-term results from the RT1 protocol. *Neuro-Oncol*. 2022;**24**(12):2200–2209. https://doi.org/10.1093/neuonc/noac124
4. Karavitaki N, Brufani C, Warner JT, *et al*. Craniopharyngiomas in children and adults: systematic analysis of 121 cases with long-term follow-up. *Clin Endocrinol (Oxf)*. 2005;**62**(4):397–409. https://doi.org/10.1111/j.1365-2265.2005.02231.x
5. Garrè ML, Cama A. Craniopharyngioma: modern concepts in pathogenesis and treatment. *Curr Opin Pediatr*. 2007;**19**(4):471–479. https://doi.org/10.1097/MOP.0b013e3282495a22
6. Lober RM, Harsh GR. A perspective on craniopharyngioma. *World Neurosurg*. 2013;**79**(5-6):645–646. https://doi.org/10.1016/j.wneu.2012.10.064
7. Cohen M, Guger S, Hamilton J. Long term sequelae of pediatric craniopharyngioma - literature review and 20 years of experience. *Front Endocrinol*. 2011;**2**:81. https://doi.org/10.3389/fendo.2011.00081
8. Khan RB, Merchant TE, Sadighi ZS, *et al*. Prevalence, risk factors, and response to treatment for hypersomnia of central origin in survivors of childhood brain tumors. *J Neurooncol*. 2018;**136**(2):379–384. https://doi.org/10.1007/s11060-017-2662-y
9. Jacola LM, Conklin HM, Scoggins MA, *et al*. Investigating the role of hypothalamic tumor involvement in sleep and cognitive outcomes among children treated for craniopharyngioma. *J Pediatr Psychol*. 2016;**41**(6):610–622. https://doi.org/10.1093/jpepsy/jsw026
10. Waber DP, Pomeroy SL, Chiverton AM, *et al*. Everyday cognitive function after craniopharyngioma in childhood. *Pediatr Neurol*. 2006;**34**(1):13–19. https://doi.org/10.1016/j.pediatrneurol.2005.06.002
11. Özyurt J, Thiel CM, Lorenzen A, *et al*. Neuropsychological outcome in patients with childhood craniopharyngioma and hypothalamic involvement. *J Pediatr*. 2014;**164**(4):876–881.e4. https://doi.org/10.1016/j.jpeds.2013.12.010
12. Mandrell BN, LaRosa K, Hancock D, *et al*. Predictors of narcolepsy and hypersomnia due to medical disorder in pediatric craniopharyngioma. *J Neurooncol*. 2020;**148**(2):307–316. https://doi.org/10.1007/s11060-020-03519-3
13. Müller HL. Increased daytime sleepiness in patients with childhood craniopharyngioma and hypothalamic tumor involvement: review of the literature and perspectives. *Int J Endocrinol*. 2010;**2010**:519607. https://doi.org/10.1155/2010/519607
14. Sateia MJ. International classification of sleep disorders-third edition: highlights and modifications. *Chest*. 2014;**146**(5):1387–1394. https://doi.org/10.1378/chest.14-0970
15. Janssens K, Amesz P, Nuvelstijn Y, *et al*. School problems and school support for children with narcolepsy: parent, teacher, and child reports. *Int J Environ Res Public Health*. 2023;**20**(6):5175. https://doi.org/10.3390/ijerph20065175
16. Tadrous R, O'Rourke D, Mockler D, Broderick J. Health-related quality of life in narcolepsy: a systematic review and

meta-analysis. *J Sleep Res*. 2021;**30**(6):e13383. https://doi.org/10.1111/jsr.13383
17. Semko J, Al Ghriwati N, Winter M, Merchant TE, Crabtree VM. Sleep-related challenges and family functioning in children and adolescents previously treated for craniopharyngioma. *J Psychosoc Oncol*. 2024;**42**(1):32–47. https://doi.org/10.1080/07347332.2023.2197433
18. Bonnet MH. Effect of sleep disruption on sleep, performance, and mood. *Sleep*. 1985;**8**(1):11–19. https://doi.org/10.1093/sleep/8.1.11
19. Berry RB, Gamaldo CE, Harding SM, *et al*. AASM scoring manual version 2.2 updates: new chapters for scoring infant sleep staging and home sleep Apnea testing. *J Clin Sleep Med JCSM Off Publ Am Acad Sleep Med*. 2015;**11**(11):1253–1254. https://doi.org/10.5664/jcsm.5176
20. Institute of Medicine (US) Committee on Sleep Medicine and Research. In: Colten HR, Altevogt BM, eds. *Sleep Disorders and Sleep Deprivation: An Unmet Public Health Problem*. Washington, DC: National Academies Press (US); 2006: http://www.ncbi.nlm.nih.gov/books/NBK19960/. Accessed January 6, 2025.
21. Lecci S, Fernandez LMJ, Weber FD, *et al*. Coordinated infraslow neural and cardiac oscillations mark fragility and offline periods in mammalian sleep. *Sci Adv*. 2017;**3**(2):e1602026. https://doi.org/10.1126/sciadv.1602026
22. Kjaerby C, Andersen M, Hauglund N, *et al*. Memory-enhancing properties of sleep depend on the oscillatory amplitude of norepinephrine. *Nat Neurosci*. 2022;**25**(8):1059–1070. https://doi.org/10.1038/s41593-022-01102-9
23. Wang H, Robinson RC, Burtnick LD. The structure of native G-actin. *Cytoskelet Hoboken NJ*. 2010;**67**(7):456–465. https://doi.org/10.1002/cm.20458
24. Pizza F, Vandi S, Iloti M, *et al*. Nocturnal sleep dynamics identify narcolepsy type 1. *Sleep*. 2015;**38**(8):1277–1284. https://doi.org/10.5665/sleep.4908
25. Vandi S, Rodolfi S, Pizza F, *et al*. Cardiovascular autonomic dysfunction, altered sleep architecture, and muscle overactivity during nocturnal sleep in pediatric patients with narcolepsy type 1. *Sleep*. 2019;**42**(12). https://doi.org/10.1093/sleep/zsz169
26. Cordani R, Veneruso M, Napoli F, *et al*. Sleep disturbances in craniopharyngioma: a challenging diagnosis. *J Neurol*. 2021;**268**(11):4362–4369. https://doi.org/10.1007/s00415-021-10794-1
27. Tachibana N, Taniike M, Okinaga T, Ripley B, Mignot E, Nishino S. Hypersomnolence and increased REM sleep with low cerebrospinal fluid hypocretin level in a patient after removal of craniopharyngioma. *Sleep Med*. 2005;**6**(6):567–569. https://doi.org/10.1016/j.sleep.2005.04.002
28. Müller HL, Gebhardt U, Teske C, *et al*. Post-operative hypothalamic lesions and obesity in childhood craniopharyngioma: results of the multinational prospective trial KRANIOPHARYNGEOM 2000 after 3-year follow-up. *Eur J Endocrinol*. 2011;**165**(1):17–24. https://doi.org/10.1530/EJE-11-0158
29. Gozal D, Wang M, Pope DW. Objective sleepiness measures in pediatric obstructive sleep apnea. *Pediatrics*. 2001;**108**(3):693–697. https://doi.org/10.1542/peds.108.3.693
30. Zhu Y, Li AM, Au CT, *et al*. Association between sleep architecture and glucose tolerance in children and adolescents. *J Diabetes*. 2015;**7**(1):10–15. https://doi.org/10.1111/1753-0407.12138
31. Toor B, Ray LB, Pozzobon A, Fogel SM. Sleep, orexin and cognition. https://karger.com/books/book/380/chapter/5563870/Sleep-Orexin-and-Cognition. Accessed October 10, 2025
32. Overeem S, van Litsenburg RRL, Reading PJ. Sleep disorders and the hypothalamus. *Handb Clin Neurol*. 2021;**182**:369–385. https://doi.org/10.1016/B978-0-12-819973-2.00025-3
33. Zaffanello M, Pietrobelli A, Cavarzere P, Guzzo A, Antoniazzi F. Complex relationship between growth hormone and sleep in children: insights, discrepancies, and implications. *Front Endocrinol*. 2024;**14**:14. https://doi.org/10.3389/fendo.2023.1332114
34. McLaughlin Crabtree V, Williams NA. Normal sleep in children and adolescents. *Child Adolesc Psychiatr Clin N Am*. 2009;**18**(4):799–811. https://doi.org/10.1016/j.chc.2009.04.013
35. Agostini A, Centofanti S. Normal sleep in children and adolescence. *Psychiatr Clin North Am*. 2024;**47**(1):1–14. https://doi.org/10.1016/j.psc.2023.06.001
36. Feinberg I, Davis NM, de Bie E, Grimm KJ, Campbell IG. The maturational trajectories of NREM and REM sleep durations differ across adolescence on both school-night and extended sleep. *Am J Physiol - Regul Integr Comp Physiol*. 2012;**302**(5):R533–R540. https://doi.org/10.1152/ajpregu.00532.2011
37. Ding L, Chen B, Dai Y, Li Y. A meta-analysis of the first-night effect in healthy individuals for the full age spectrum. *Sleep Med*. 2022;**89**:159–165. https://doi.org/10.1016/j.sleep.2021.12.007
38. Sprajcer M, Gupta C, Roach G, Sargent C. Can we put the first night effect to bed? An analysis based on a large sample of healthy adults. *Chronobiol Int*. 2022;**39**(12):1567–1573. https://doi.org/10.1080/07420528.2022.2133611
39. McCall C, McCall WV. Objective vs. subjective measurements of sleep in depressed insomniacs: first night effect or reverse first night effect? *J Clin Sleep Med JCSM Off Publ Am Acad Sleep Med*. 2012;**8**(1):59–65. https://doi.org/10.5664/jcsm.1664
40. Moser D, Anderer P, Gruber G, *et al*. Sleep classification according to AASM and Rechtschaffen & Kales: effects on sleep scoring parameters. *Sleep*. 2009;**32**(2):139–149. https://doi.org/10.1093/sleep/32.2.139
41. Witcraft SM, Wickenhauser ME, Russell KM, *et al*. A latent profile analysis of sleep, anxiety, and mood in youth with craniopharyngioma. *Behav Sleep Med*. 2022;**20**(6):762–773. https://doi.org/10.1080/15402002.2021.1996363

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